Empirical Formula of Magnesium Oxide

Grade 11 Chemistry | Practical worksheet

Name: _____ Date: _____ Lab partner: _____

Aim and background

Aim: To find the empirical formula of magnesium oxide by heating a known mass of magnesium in air and measuring the mass of oxide formed.

When magnesium is heated strongly in air it reacts with oxygen to form a white solid, magnesium oxide. The mass gained by the crucible contents equals the mass of oxygen that has combined with the magnesium. Converting both masses to moles gives the simplest whole-number ratio of Mg to O: the **empirical formula**.

Safety and equipment

Safety: Wear goggles throughout. Do not look directly at burning magnesium, because the bright flame gives off UV light. The crucible, lid and pipe-clay triangle stay hot for several minutes, so handle them only with tongs. Do not breathe in the white MgO smoke.

Equipment	Uncertainty
Electronic balance (2 d.p.)	± 0.01 g
Crucible with lid, tongs	–
Pipe-clay triangle, tripod, heatproof mat, Bunsen burner	–
Magnesium ribbon, about 15 cm (about 0.2–0.3 g), and emery paper	–
Dropping pipette and distilled water	–

Method

1. Heat the empty crucible and lid strongly for 2 minutes to drive off moisture. Let them cool on the mat, then weigh them together.
2. Clean the magnesium ribbon with emery paper until it is shiny, removing the oxide layer. Coil it loosely, put it in the crucible, and weigh the crucible, lid and Mg.
3. Set the crucible on the pipe-clay triangle with the lid on. Heat gently at first, then strongly.
4. When the Mg starts to glow, use tongs to lift the lid slightly for a moment every 30 seconds. This lets oxygen in while keeping MgO smoke from escaping.
5. When the Mg no longer glows, take the lid off and heat strongly for 3 more minutes.
6. Let the crucible cool. Add about 5 drops of distilled water to change any magnesium nitride (Mg_3N_2) into the hydroxide. Put the lid back on and heat gently, then strongly, for 2 minutes.
7. Let the crucible cool and weigh it with the lid and product.
8. Heat for 2 more minutes, let it cool, and weigh again. Repeat until two masses in a row agree to within 0.01 g (constant mass).

Raw data

Record every mass to 2 decimal places, matching the balance's precision.

Measurement	Mass / g (± 0.01 g)
A: crucible + lid (empty)	
B: crucible + lid + Mg	
C1: crucible + lid + product, 1st weighing	
C2: 2nd weighing	
C3: 3rd weighing (if needed)	
C: final constant mass	

Qualitative observations (colour, flame, smoke, appearance of product):

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Processing the data

Use M(Mg) = 24.31 g mol^{-1} and M(O) = 16.00 g mol^{-1} from the data booklet. Treat their uncertainties as negligible.

Uncertainty rules: When you add or subtract, add the **absolute** uncertainties. When you multiply or divide, add the **percentage** uncertainties.

Step	Quantity	Working	Value	Absolute unc.	% unc.
1	mass of Mg / g	B − A		± 0.02 g	
2	mass of MgO / g	C − A		± 0.02 g	
3	mass of O / g	C − B		± 0.02 g	
4	n(Mg) / mol	mass of Mg ÷ 24.31			same as step 1
5	n(O) / mol	mass of O ÷ 16.00			same as step 3
6	mole ratio O : Mg	n(O) ÷ n(Mg)			step 4 + step 5

To convert % to absolute uncertainty: absolute unc. = (% unc. ÷ 100) × value

Mole ratio O : Mg = _____ ± _____ Empirical formula = _____

Percentage error compared with the accepted ratio of 1.00 = _____ %

Percentage yield. Use the equation $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$ and M(MgO) = 40.31 g mol^{-1}. Magnesium is the limiting reactant, so it sets the theoretical yield.

Step	Quantity	Working	Value	Absolute unc.	% unc.
7	theoretical mass of MgO / g	n(Mg) × 40.31			same as step 1
8	actual mass of MgO / g	from step 2		± 0.02 g	
9	percentage yield / %	(step 8 ÷ step 7) × 100			step 7 + step 8

Percentage yield = _____ ± _____ %

Conclusion and evaluation

1. Does the accepted ratio of 1.00 fall inside your uncertainty range? What does this tell you about whether your error is random or systematic?

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2. Can a percentage yield above 100% happen in this experiment? Suggest one cause.

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3. Which mass has the largest percentage uncertainty? Suggest two ways to reduce it.

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4. Which reactant is limiting and which is in excess? Explain how lifting the lid during heating keeps it that way, and predict the effect on percentage yield if the lid were never lifted.

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5. Complete the table: does each error make the O : Mg ratio and the percentage yield **too high** or **too low**? Explain why.

Error	O : Mg ratio (high / low)	% yield (high / low)	Reason
(a) MgO smoke escapes when the lid is lifted			
(b) Some magnesium does not react			
(c) Some Mg forms Mg_3N_2 instead			
(d) Ribbon not cleaned before weighing			

6. **Extension:** Mg_3N_2 reacts with water to give $Mg(OH)_2$ and ammonia, and $Mg(OH)_2$ decomposes on heating to MgO. Write both equations, and explain how step 6 of the method improves accuracy.

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